PROJECT VISION DECK

AutoRAG

Automated RAG Optimization Pipeline

End-to-end agent that builds, evaluates, improves, and deploys RAG systems automatically.

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What this project is about

AutoRAG is designed to solve one painful problem:

Building a good RAG system is slow, manual, and guesswork-heavy.

🔍 What is a RAG system?

- "Chat with your documents"
- Internal company knowledge bots
- AI assistants that answer using your data

😩 The Problem Today

- Manual tuning of chunking, embeddings, retrieval, prompts
- No clear scoring system
- No automation
- No reliable way to know “this is good”

It works like this:

1. Search relevant documents

2. Feed them to an AI model

3. Generate an answer grounded in those documents

👉 **Engineers repeat: Try → test → tweak → repeat (for days)**

🚀 What AutoRAG Does

AutoRAG turns that manual process into an automated loop

1. Builds a RAG pipeline →
2. Evaluates how good it is →
3. Finds what's wrong →
4. Improves it automatically →
5. Repeats until it gets better

🔁 **The core idea (very important): AutoRAG is a closed-loop system**

Build → **Evaluate** → **Diagnose** → **Improve** → **Repeat**

✅ **No manual tuning required.**

How It Works Internally

It uses multiple **6 agent cards**, each with a job:

Ingestion

Prepares your documents

Builder

Creates the RAG pipeline

Evaluator

Tests it with auto-generated questions

Diagnoser

Finds why it failed

Improver

Tries better configurations

Deployer

Puts the best version into production

📊 How It Measures Quality

🎯 The Core Scoring Formula

$$\text{Score} = 0.25 \cdot \text{Retrieval} + 0.35 \cdot \text{Quality} + 0.25 \cdot \text{Faithfulness} - 0.10 \cdot \text{Latency} - 0.05 \cdot \text{Cost}$$

🔍 Retrieval

25%

Did it fetch the right information?

✍️ Quality

35%

Is the answer actually good? Clarity, completeness, relevance.

✅ Faithfulness

25%

Is it grounded in documents? No hallucinations.

⚡ Latency

−10%

How fast is the response? Slower pipelines are penalized.

💰 Cost

−5%

How expensive is it to run?

Score: 0.85+ → Production-ready **0.70–0.84 → Needs improvement** **< 0.70 → Poor quality**

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💡 Why This Matters

AutoRAG removes the biggest bottleneck in AI systems:

❌ BEFORE

- ❌ Manual experimentation
- ❌ Need for labeled data
- ❌ Trial-and-error tuning

→

✅ AFTER

- ✅ Automated optimization
- ✅ Measurable quality
- ✅ Fast iteration (minutes instead of days)

🧑‍💻 Who It's For

- Engineers building AI products
- Data scientists running experiments
- Companies deploying internal knowledge bots
- Developers who want fast, reliable RAG systems

One-Line Summary

AutoRAG is an automated system that builds and continuously improves AI document-based question-answering systems without manual tuning or labeled data.

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📊 The Scoring System — Deep Dive

The scoring in AutoRAG decides: "Is this RAG pipeline good or bad?"

🔍 Retrieval (25%)

Did the system fetch the right information? Checks if relevant document chunks were retrieved. If retrieval is bad, everything else fails.

✍️ Quality (35%) — Most important

Is the answer actually good? Clarity, completeness, relevance. Measured using an LLM acting as a judge.

✅ Faithfulness (25%)

Is the answer grounded in the documents? No hallucinations, no made-up facts. Critical for enterprise trust.

⚡ Latency (–10%)

How fast is the response? Slower pipelines are penalized. Encourages real-world usability.

💰 Cost (–5%)

How expensive is it to run? API usage (LLMs, embeddings), infrastructure cost.

📊 Final Score Behavior

Score: 0.0 → 1.0   Higher = better pipeline
0.85 → Production-ready
0.70 → Needs improvement
0.50 → Poor quality

🔁 How it's used in the system

→ Evaluator → calculates it

→ Diagnoser → explains why it's low

→ Improver → tries to increase it

→ Experiment Tracker → compares runs

📃 Simple way to think about it: Score = How correct + how truthful + how useful – how slow – how expensive

🧪 Real Example: How a Pipeline Gets Scored

🧪 Scenario: RAG system on company HR documents

User question: “What is the maternity leave policy?”
Ground truth: “Employees are entitled to 26 weeks of paid maternity leave.”

Component	Raw Score	Weight	Contribution
🔍 Retrieval	0.70	×0.25	+0.175
✍️ Quality	0.65	×0.35	+0.2275
✅ Faithfulness	0.60	×0.25	+0.15
⚡ Latency penalty	0.20	×0.10	−0.020
💰 Cost penalty	0.10	×0.05	−0.005

✅ Final Score ≈ 0.53 → Poor quality → System triggers improvement

🔁 After Improvement

New answer: “Employees are entitled to 26 weeks of paid maternity leave.”

Retrieval: **0.85**
Quality: **0.90**
Faithfulness: **0.95**

Final Score ≈ 0.82 ✅ Production-ready

🧠 Diagnosis: Partial Answer + Mild Hallucination → 🔧 Actions: Increase top_k, stricter prompt, lower temperature

🧾 Key Takeaway

This scoring system doesn't just say "right or wrong."

- 🔍 Did we retrieve the right info?
- ✍️ Did the model answer well?
- 🚫 Did it hallucinate?
- ⚡ Is it fast and cheap enough?

👉 And then uses that to automatically improve the pipeline

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